

Antibiotic 'removal' is not detoxification

- **310 ng/g tetracycline** was measured in Lagos hospital wastewater sludge (Ajibola & Zwiener 2022)
- **tetA / tetB / tetM** tetracycline-resistance genes circulate in Lagos aquatic isolates (2025 field study)
- **Yet:** parent-molecule disappearance is counted as success — while products and selection continue

310.2

ng/g TC · Lagos sludge

<0.05–8.84

µg/L range · Lagos waters (class)

3

resistance classes in environment

ng–µg/L

sub-MIC selection pressure persists

The molecule can vanish from the LC-MS trace while its active epimer remains — and sub-inhibitory residues keep training resistant bacteria.

Nigeria needs remediation verified **SAFE**, not just analytically invisible.

Three facts the routine metrics miss

REMOVAL $\neq$ DEGRADATION

Sorption & epimerisation hide the molecule without changing its activity.

DEGRADATION $\neq$ DETOXIFICATION

4-epitetracycline retains antibacterial activity;
anhydrotetracycline alters it.

DEGRADERS CAN BE RISKS

The best tetracycline-destroying enzymes ARE resistance genes (tet(X) family).

CONTAMINANT



PARENT DISAPPEARS



...PRODUCT?



...ACTIVITY?



...AMR SELECTION?

Analytics answers the first box. TETRA-SHIELD computes all five.

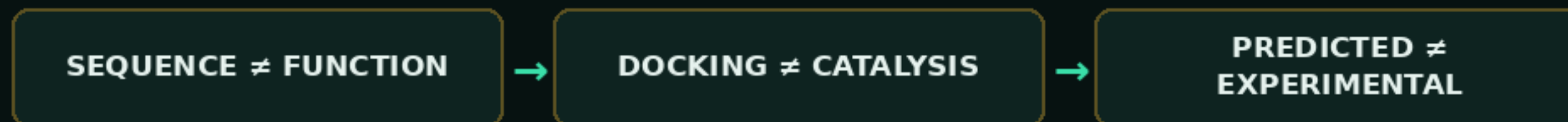
Nature already runs this chemistry

- **Tet(X) family:** FAD monooxygenases hydroxylate tetracycline at C11a → Mg^{2+} -chelating pharmacophore destroyed → lost antibacterial activity
- **Fungal white-rot enzymes:** multicopper oxidases + Mn-peroxidases oxidise tetracyclines (Suda 2012; Wen 2010)
- **Insect microbiomes:** black-soldier-fly gut microbiomes enrich laccase/oxidase-like genes and triple degradation (2023–2025 studies)



Tension: Tet(X) is literature's strongest degrader — and a mobile resistance determinant.

Deployment must therefore prefer purified, cell-free enzymes — never released genes/organisms.



Can this transformation help — and could it create another problem



- **EXPLAINABLE:** Every score: positive contributors vs risk penalties, click 'WHY?'
- **STANCE-AWARE:** WHAT-IF presets recalculate: safety-first, AMR-conservative, low-cost...
- **OFFLINE-FIRST:** Judges' demo runs with zero network; all data provenance-tagged
- **HONEST:** UNKNOWN stays UNKNOWN; negatives are reported, never hidden

THE ONE-SENTENCE DIFFERENTIATION

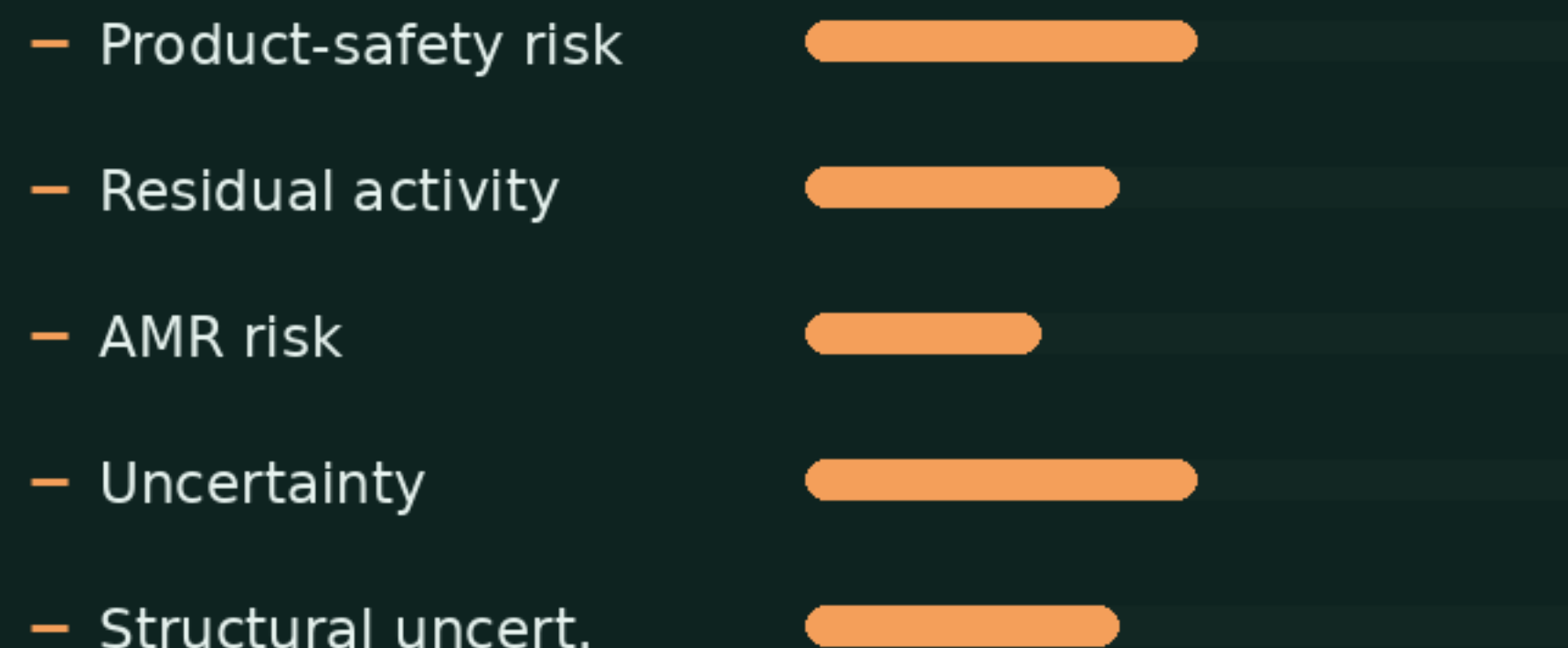
BLAST answers 'similar?'. Docking answers 'binds?'. CARD answers 'resistance?'.
TETRA-SHIELD chains them into one decision: which biocatalyst to validate first — and what ha

13 inspectable components — no black box

POSITIVE EVIDENCE (8)



RISK PENALTIES (5)



$$\text{score} = 100 \times (\sum w^+ \cdot \text{evidence} - 0.85 \cdot \sum w^- \cdot \text{risk} + 0.30)$$

Weights are sliders, not doctrine — and stability is measured, not assumed.

No ML fitted on n=6 — deterministic evidence integration (choice documented in MODEL_CARD).

Actually computed. Actually validated.

VALIDATED DOCKING

-9.7

kcal/mol · tetracycline ↔ TetX2 (2Y6R+FAD)

1.84 Å

re-dock RMSD vs co-crystal ligand → PASS

smina/Vina 1.1.2 · seed 42 · exhaustiveness 16

CONTACT MAP (TOP POSE)

HIS234

ARG213

GLU114

PHE224

MET215

PR0318-GLY321

FAD 2.36 Å · matches literature pocket

HONEST NEGATIVES

Laccase rigid docking: all poses
≈ 0 kcal/mol — reported, not
hidden. Substrate case rests on
E2 literature.

DOCKING ≠ CATALYSIS

Sequence intelligence (real alignments):

TetX ↔ TetX2 99.5%



↔ Tet(X3) 82%



↔ fungi 18-21%



2 independent solutions

The safety layers change the decision

AMR SHIELD

TetX family	HIGH
TetX2	HIGH
Tet(X3)	HIGH
Laccase	LOW
MnP1	LOW
BSFL family	MODERATE

PRODUCT SAFETY TRIAGE

Epimer: RETAINS activity — the removal trap.

Anhydro-TC: altered activity + TetX inhibitor.

11 α -OH product: unstable → label UNCERTAIN, not 'safe'.

CASSAVA LAYER (EVIDENCE-BOUNDED)

Sludge biochar adsorbs TC: 154.45 mg/g (batch).

Peel biochar: CFX evidence only — labelled as such.

ADS ORPTION $\neq$ destruction; spent-material fate modelled.

CASSAVA WASTE



BIOCHAR



PRE-CONCENTRATE



CELL-FREE ENZYME



SAFETY CHECK



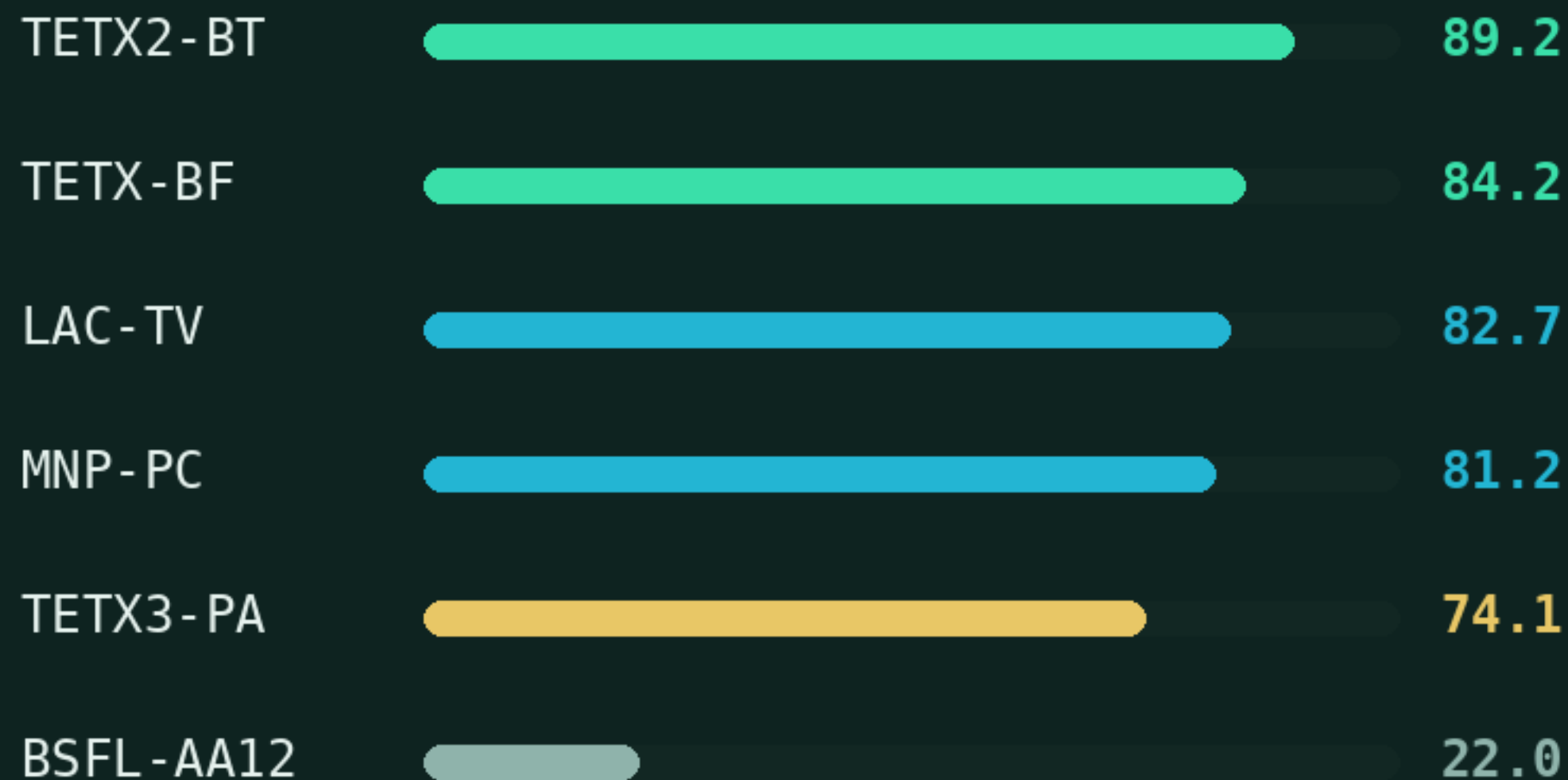
END-OF-LIFE

Recommended architecture — Option C: adsorption pretreatment + enzymatic transformation.

Never merely moving pollution from water to solid waste.

Rankings that survive being poked

BALANCED PRIORITY RANKING



TetX2 leads — with HIGH-CONCERN AMR caveat printed on its own card.

0.955

Kendall's W · 400 weight sets

39/39

automated tests pass

ABLATION — INTEGRATION MATTERS

Remove AMR layer → resistance enzymes inflate to ~100/100. Remove product safety → fungal/TetX gap inverts. Effects visible, not hidden.

STANCE-AWARE (WHAT-IF)

AMR-conservative stance: LAC-TV takes #1 (86.0), resistance enzymes demoted to #3–5. Decision support, not arbitrary AI ranking.

Frugal by design — because the deployment context demands it

- **Software:** runs offline on a laptop; venue-proof demo; zero cloud dependency
- **Materials:** feedstock is a free waste stream in the world's largest cassava producer
- **Biocatalyst form:** purified cell-free enzyme — no GMO release, no gene transfer
- **Pilot sites:** hospital + farm effluent cartridges → municipal polishing

PHASE 1

COMPUTE (done)

PHASE 2

LAB VALIDATION

PHASE 3

EFFLUENT PILOT

PHASE 4

MUNICIPAL SCALE

Revenue paths: engine licensing · enzyme-cartridge product with local biochar manufacturing · circular-economy grants.

Costs not quoted on purpose: they require the wet-lab phase the roadmap defines.

From decision engine to verified clean water

STEP 1

express & purify candidate enzyme

STEP 2

tetracycline degradation assay (HPLC)

STEP 3

LC-MS/MS product identification

STEP 4

residual antimicrobial activity assay

STEP 5

toxicity + AMR monitoring

STEP 6

cassava-biochar cartridge pilot

Platform next modules (only where evidence exists):

DOXY-SHIELD



CIPRO-SHIELD



MYCO-SHIELD (mycotoxins)



PESTI-SHIELD



Removal is not detoxification. TETRA-SHIELD makes remediation safe, explainable, and ready for